

Software Features and Vendors

Software	Features	Vendor
SignalVu	Vector signal analysis, pulse analyser and spectrum analyser functionality on an oscilloscope	Tektronix
LabVIEW	Data acquisition, instrument control, test automation, analysis and signal processing	National Instruments
Certus 2.0	Automatically measures all the signals likely to be needed in each FPGA, in a multi-FPGA ASIC prototype	Tektronix
Measurement Studio	Data acquisition, instrument control, test automation, analysis and signal processing integrated in MS Visual Studio environment	National Instruments
AX-Series	ATE for AXIe and PXI modules, characterisation and production testing of consumer-RF and mixed-signal semiconductors	Aeroflex

their IP, and community members who do not want to share their IP for free. T&M players, therefore, have to come up with innovative ways to incentivise content sharing in the ecosystem.

T&M players have to ensure that their hardware and software platforms are standardised, open and customisable enough for the ecosystem to develop. At the same time, they have to ensure the quality of their products and their compatibility with these additions. Active industry groups such as the IVI Foundation, PXI Systems Alliance and LXI Consortium are bringing industry players together by standardising the hardware platforms. By developing a common means of communicating to similar instruments across multiple vendors at the application programming interface level, the IVI Foundation has reduced the learning curve for users and the development cycle for vendors. This has opened the door for third parties to create drivers, aggregation websites to house them, and abstraction layers to be created on top of them—essentially helping the ecosystem grow even further.

Naresh Kumar, general manager, Application Engineering Organisa-

tion, Agilent Technologies India, says, "Some of the challenges faced while developing a suitable ecosystem for the products include interworking with standard software development platforms and languages such as C, C++, C#, VB, VB .NET, VEE or LabVIEW, along with test automation-specific languages like Agilent's VEE Pro and Test Exec SL, besides developing instrument drivers for multiple platforms (IVI, MATLAB, LabVIEW). Another interesting challenge is with regard to integrating reporting features for generating test reports into the test automation software."

The future of software-centric ecosystems

There should be no doubt about the fact that software-centric solutions are here to stay. Without such an ecosystem, many viable open platforms have struggled. The xTCA platforms had seen adoption in telecommunication infrastructures and interest from the high-energy physics community, but they failed to develop a strong ecosystem in automated testing. The multiple form factor, communication bus and software options presented by the

platform have delayed or complicated adoption by leading vendors. While efforts to rein in those options and improve them for automated testing are under way at the AXIe Consortium, success or failure will be dictated by the use of a software-centric ecosystem.

Over the next three to five years, automated test systems will become more software-centric and ecosystems will have more of an impact on the value users derive from these platforms. In the past, your test system was only as valuable as the investment of time and money that you made in it. Going forward, your system will benefit from the entire community of third-party suppliers, integrators, consultants, and derived standards supporting the software ecosystem at its core. This is a crucial element in meeting the demands of next-generation device testing. Software vendors taking greater advantage of their ecosystems and leveraging commercialisation models for third-party IP will have a transformative effect on the test and measurement industry.

Naresh Kumar adds, "The trend is moving towards an emphasis on the seamless integration of T&M-specific software with standard software languages or platforms, thereby providing an opportunity for easy integration of software and hardware. In the near future, there will still be a need for further development in this domain due to the requirement for ease of automation in the manufacturing and design domains, which will drive developments in the software ecosystems enabled by T&M players like Agilent." ●

The author is a tech correspondent at EPY Bengaluru

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